HARISANKAR NAGARAJAN

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PROFESSIONAL SUMMARY

Results-driven Mechanical Engineer with 9+ years of experience in product development, value engineering, component engineering, mechanical design, and manufacturing engineering across renewable energy, semiconductor, automotive, and industrial products. Proven expertise in NX, SolidWorks, CREO, Teamcenter, SAP, GD&T, DFM/DFA, DFMEA. Experienced in leading end-to-end product development from concept design through manufacturing support, engineering change management, cost optimization, component standardization, product qualification, regulatory standards, and project scheduling and product lifecycle management while delivering cost-effective, manufacturable, and high-quality engineering solutions.

CORE COMPETENCIES

- Mechanical Design & Product Development
- Component Engineering & Material Selection
- Design Verification & Design Validation
- Product Structure & BOM Management
- 3D CAD Modeling & 2D Drafting
- Tolerance Stack-up analysis, GD&T, & Engineering Change Management
- Root Cause Analysis, Prototype Validation, & Production Support
- DFM/DFA & Assembly Process Development
- Tooling, Jigs & Fixture Design, Manufacturing Process Optimization
- Cross-functional Stakeholder Coordination
- Lean Manufacturing & Continuous improvement

PROFESSIONAL EXPERIENCE

Senior Technical Professional-I | Capgemini Engineering Services | Nov 2022 – Present
Client: Siemens Energy

- Led & participated in design reviews, engineering change reviews, and technical discussions with global engineering teams to ensure compliance with manufacturing and quality requirements.
- Conducted quality governance meeting with internal stake holders to monitor project performance and Managed project schedules, supported strategic decision-making through risk and SWOT analysis.

- Designed and supported mechanical assemblies including fabricated structures, welded structures, tooling systems, and production equipment with emphasis on manufacturability and assembly validation.
- Supported integration of Generator assemblies, rotating equipment, sensors, and mechanical components during design validation and manufacturing support.
- Created and released 3D models, 2D manufacturing drawings, and engineering documentation in accordance with design standards and project specifications.
- Prepared and maintained Engineering BOM (EBOM) and supported Manufacturing BOM (MBOM) creation in SAP and Teamcenter to enable manufacturing readiness and product lifecycle management.
- Evaluated engineering changes for cost, manufacturability, and quality impacts.
- Integrated mechanical components, sensors, and auxiliary hardware into product assemblies while ensuring compliance with functional and manufacturing requirements.
- Supported assembly validation and design qualification, product validation during prototype and production builds by resolving mechanical fitment and assembly-related issues.
- Developed manufacturing work instructions, assembly procedures, and engineering documentation to improve production readiness and support product industrialization.
- Provided technical guidance and knowledge sharing to junior engineers during design reviews and project execution.

Lead Engineer | HCL Technologies Ltd | Jun 2021 – Nov 2022
Client: Applied Materials

- Led end-to-end obsolescence management projects using NX and Teamcenter.
- Conducted DFM/DFA reviews to reduce product complexity through design optimization, improve manufacturability and assembly feasibility.
- Performed design-to-cost evaluations and Supported cost optimization through component standardization.
- Provided mechanical support throughout product lifecycles including design releases, ECOs (Engineering Change Orders), and production support activities.
- Prepared engineering release packages, BOMs, drawings, managed document revisions and version control, and supported ECR/ECN/ECO documentation for production release through Teamcenter & SAP and Coordinated engineering change implementation across global teams.
- Evaluated and selected mechanical components, materials, and standard hardware based on functional, manufacturing, quality, and cost requirements.
- Participated in structured problem-solving activities to address manufacturing and product lifecycle challenges.
- Standardized engineering documentation and Performed design simplification to reduce product cost.
- Supported manufacturing teams with process improvements and technical issue resolution.
- Developed Best Known Method (BKM) documentation and supported COE activities.

- Evaluated alternative manufacturing processes for cost reduction.

NPD Engineer | Pennar Industries Ltd | Mar 2019 – Jul 2020
Clients: Ashok Leyland, Doosan Bobcat, VRV, ICF, Kone Elevators, Fujitec,

- Acted as the design and process authority for fabrication products, molding products, press tools, injection molds, jigs, fixtures, and automotive components.
- Designed mechanical assemblies, machine fixtures, and production equipment to improve manufacturing efficiency and support new product introduction.
- Led value engineering initiatives by optimizing designs, selecting cost-optimized materials and manufacturing processes, and implementing engineering changes, resulting in a 10% reduction in manufacturing costs while maintaining product quality, performance, and reliability.
- Converted customer and manufacturing requirements into design specifications and validated critical functional characteristics.
- Performed cost-benefit analysis before implementing engineering changes.
- Supported manufacturing feasibility and cost optimization through material selection and process evaluation.
- Collaborated with suppliers to identify cost-saving opportunities and Improved manufacturability while maintaining product quality.
- Led technical design verification, material selection activities, and actively contributed to FMEA sessions to identify and mitigate potential design risks.
- Conducted design verification through tolerance analysis, manufacturability reviews, prototype evaluation, and engineering validation.
- Performed engineering calculations, tolerance stack-up analysis, and DFM reviews to ensure smooth transition of designs into production.
- Conducted design risk assessments through DFMEA and implemented mitigation plans to improve product reliability.

NPD Engineer | Precision Pressform Industries | Jul 2015 – Mar 2019
Clients: Valeo, Schaeffler India, Lucas TVS, Repco

- Converted customer requirements into engineering specifications and design solutions through close collaboration with stakeholders.
- Led design and development of automotive sheet metal stamping components, press tools, jigs, and fixtures from concept through production release.
- Conducted tooling feasibility studies, process planning, optimized press tool layouts, and implemented manufacturing process optimization initiatives, improving productivity through process optimization, reducing waste, and enhancing product quality.
- Reduced manufacturing cost through tooling modifications.
- Optimized tooling cost through design improvements & Conducted supplier cost reviews for tooling and components.
- Evaluated manufacturing tooling and machine performance, implemented engineering improvements, and verified effectiveness through process capability studies.

- Performed engineering calculations including tolerance stack-up, material selection, fastener sizing, and structural design validation.
- Conducted value analysis workshops to identify cost-out opportunities, work plans, and schedules, successfully delivering projects within scope, budget, and timeline.
- Participated in manufacturing process reviews to ensure compliance with safety, EHS, and operational, Regulatory standards during industrialization activities.
- Prepared & released process documentation APQP, PFMEA, PFD, SOP, and Control Plan.
- Managed tooling and process validation for high-volume sheet metal components, ensuring adherence to IATF 16949:2016.

KEY ACHIEVEMENTS

- Implemented lean practices that cut non-value-added engineering activities by 20%.
- Reduced project cycle time by 15% through structured design verification, engineering standardization, and Cross-functional team coordination.

TECHNICAL SKILLS

- CAD & Design: NX, SOLIDWORKS, CREO, AutoCAD
- PLM & ERP: Teamcenter, Windchill, SAP, Product Lifecycle Management (PLM)
- Engineering: GD&T (ASME Y14.5), ISO 2768, Mechanical calculations, Tolerance Stack-up, Design Verification & Validation.
- Quality tools: DFM/DFA, APQP, PPAP, DFMEA, & PFMEA, Control Plan, SPC, 5 Why, Fishbone, 8D
- Manufacturing: Sheet Metal Fabrication, Tooling Design, Machining, Assembly Processes, Manufacturing Process Optimization, Production Support, Production readiness
- Project tools: JIRA, Power BI
- DITA: Vismockup, Adobe photoshop, Tech pub studio, Cortona

CERTIFICATIONS

- IATF 16949:2016 Internal Auditor (Automotive QMS System)
- Advanced Product Quality Planning (APQP)
- Lean six sigma green belt (Pursuing)

EDUCATION

Bachelor of Engineering (B.E.) – Mechanical Engineering | 2011-2015

Sri Shanmugha College of Engineering & Technology